MANAGERIAL ECONOMICS

Exam Study Guide

International MBA – Core II

IE University – IE Business School

Professor Daniel Fernández Kranz, Ph.D.

Sessions 1–11 | Microeconomics · Macroeconomics · International Economics

BM

Supply & Demand · Elasticity · Consumer & Producer Surplus · Price Controls · Tax Incidence
GDP & CPI · Fiscal & Monetary Policy · AS-AD Model · Phillips Curve
Comparative Advantage · Tariff Analysis · Industrial Policy · Exchange Rates · Balance of Payments

Case Studies: FrescoBrew Analytics · Uber Consumer Surplus · Canada Carbon Tax · Japan Macro Analysis

Tools: OLS Regression · Instrumental Variables · AS-AD Diagrams · Laffer Curve

March 2026

Contents

1 Module 1: Free Markets and Their Limits (Session 1) 3

1.1 The Invisible Hand and the Case for Free Markets . . . 3

1.2 When Markets Fail . . . 3

1.3 Government Responses to Market Failures . . . 3

1.4 The Organ Market Debate: A Framework . . . 4

2 Module 2: Supply, Demand, and Elasticity (Sessions 2–3) 5

2.1 The Demand Curve . . . 5

2.2 The Supply Curve . . . 5

2.3 Market Equilibrium . . . 5

2.4 Price Elasticity of Demand . . . 5

2.5 Cross-Price and Income Elasticity . . . 6

2.6 OLS Regression and Estimating Elasticity (FrescoBrew Case) . . . 6

2.6.1 The FrescoBrew Three-Estimate Comparison . . . 7

3 Module 3: Welfare, Price Controls, and Minimum Wage (Session 4) 8

3.1 Consumer Surplus, Producer Surplus, and Social Surplus . . . 8

3.2 The Uber Case: Measuring Consumer Surplus . . . 8

3.3 Price Ceilings and Price Floors . . . 8

3.3.1 Price Ceiling (Maximum Price) . . . 9

3.3.2 Price Floor (Minimum Price) . . . 9

3.4 The Minimum Wage Debate . . . 9

4 Module 4: Tax Incidence and the Laffer Curve (Session 5) 10

4.1 Average Effective Tax Rate (AETR) . . . 10

4.2 Marginal vs. Average Tax Rates . . . 10

4.3 Tax Incidence Theory . . . 10

4.4 Deadweight Loss (DWL) of Taxes . . . 11

4.5 The Laffer Curve . . . 11

4.6 Canada's Carbon Tax + Income Rebate (CT+IR) . . . 12

5 Module 5: Measuring Economic Activity – GDP and CPI (Session 6) 13

5.1 Gross Domestic Product (GDP) . . . 13

5.2 Nominal vs. Real GDP . . . 13

5.3 PPP-Adjusted GDP for Cross-Country Comparisons . . . 13
5.4 Extensive vs. Intensive Growth . . . 14
5.5 GDP, Growth, and Inequality – The GINI Coefficient . . . 14
5.6 The Consumer Price Index (CPI) . . . 14
5.7 Real vs. Nominal Interest Rates . . . 15

6 Module 6: Fiscal and Monetary Policy (Sessions 7–8) 16
6.1 The AS-AD Framework . . . 16
6.2 Four Macroeconomic Scenarios . . . 16
6.3 The Phillips Curve . . . 16
6.4 Policy Toolkit . . . 17
6.4.1 Fiscal Policy . . . 17
6.4.2 Monetary Policy . . . 17
6.5 Arguments Against Active Stabilization Policy . . . 17
6.6 Government Debt Dynamics . . . 18
6.7 Historical Episodes . . . 18

7 Module 7: International Trade and Tariffs (Session 9) 19
7.1 Comparative Advantage . . . 19
7.2 Gains and Losses from Trade (Neoclassical Analysis) . . . 19
7.3 The China Shock (Autor et al., 2016) . . . 20
7.4 Tariff Analysis . . . 20

8 Module 8: Industrial Policy in an Open Economy (Session 10) 21
8.1 The Infant Industry Argument . . . 21
8.2 Import Substituting Industrialization (ISI) vs. Export Oriented Industrialization (EOI) . . . 21

9 Module 9: Exchange Rates and Balance of Payments (Session 11) 22
9.1 Exchange Rate Regimes . . . 22
9.2 The Balance of Payments . . . 22
9.3 Determinants of Exchange Rates . . . 22
9.4 Fiscal and Monetary Policy Interactions with Exchange Rates . . . 23
9.5 The US Dollar Outlook . . . 23

10 Additional Key Concepts (Professor's Exam Hints) 24

11 Quick Reference Formula Sheet 25

12 Practice Problems with Worked Solutions **26**

Module 1: Free Markets and Their Limits (Session 1)

The Invisible Hand and the Case for Free Markets

Adam Smith's central insight is that a free market economic system channels individual self-interest toward meeting the needs of society. A **price** serves two roles simultaneously: it is a *signal* (conveying information about scarcity) and an *incentive* (motivating producers and consumers to act). This "invisible hand" mechanism means that no central planner is required to coordinate production—the profit motive does the work.

Why This Matters for Managers

Most business decisions happen within markets. Understanding when markets work well—and when they fail—is the foundation for evaluating government policy, regulation, and competitive strategy.

When Markets Fail

Markets fail when prices do not reflect the true cost or value of goods. Three main categories:

1. Externalities. A cost or benefit imposed on third parties not involved in the transaction. Climate change (negative externality) and education (positive externality) are classic examples. When externalities exist, the market produces too much of goods with negative externalities and too little of goods with positive externalities.

2. Asymmetric Information. One party knows more than the other. Examples: the financial crisis (risk of instruments), opioid crisis (addictiveness), used cars (quality), carbon offsets (quality). This leads to *adverse selection* (bad products drive out good) and *moral hazard* (hidden actions after a deal).

3. Market Power (Monopoly/Oligopoly). When a single firm or small group can influence prices, output is restricted below the socially optimal level and prices are set above marginal cost.

Government Responses to Market Failures

Three types of intervention:

1. **Markets are forbidden** – e.g., organ markets, child labor. Sometimes government provision replaces the market (public education, kidneys via donation systems).
2. **Ex-post penalties** – fines and lawsuits after harm occurs (financial fraud, pharmaceutical side effects).
3. **Ex-ante regulation** – rules imposed before harm (banking capital requirements, drug approval processes).

Excessive regulation can itself harm growth. Dawson & Seater (2013) found that regulation (measured by pages in the Code of Federal Regulations) had a significant negative effect on US firm productivity over 50 years. **Regulatory capture** – when regulators act in the interest of the industry they regulate rather than the public – is a constant risk.

The Organ Market Debate: A Framework

The kidney market debate illustrates the fundamental tradeoff. Iran is the only country allowing kidney sales and has virtually eliminated its waiting list. The argument *for*: prices create incentives that increase supply, reducing the 17 daily deaths in the US from organ shortage. The argument *against*: markets can "crowd out" intrinsic values—paying for organs may commodify the human body and exploit the poor. This debate exemplifies the tension between efficiency (market solutions) and equity/ethics.

Module 2: Supply, Demand, and Elasticity (Sessions 2–3)

The Demand Curve

The demand curve shows the quantity of a good consumers are willing and able to buy at each price, holding all other factors constant (*ceteris paribus*). It slopes downward: as price rises, quantity demanded falls.

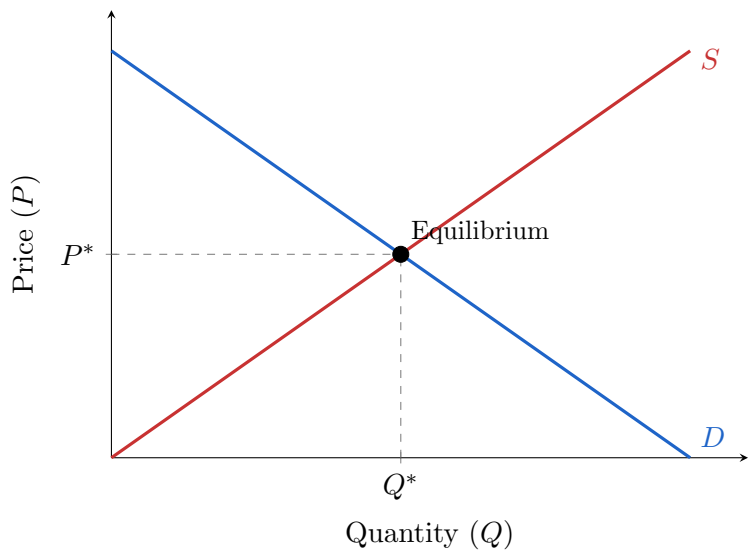
Demand shifters (factors that move the entire curve): income, consumer preferences, prices of related goods (substitutes and complements), expectations, population, advertising, weather/seasonality.

The Supply Curve

The supply curve shows the quantity producers are willing to sell at each price. It slopes upward: higher prices incentivize more production. **Supply shifters**: input costs, technology, number of sellers, expectations, regulations, taxes.

Market Equilibrium

Equilibrium occurs where supply equals demand. At the equilibrium price P^*, the quantity supplied equals the quantity demanded (Q^*). Any price above P^* creates a surplus (excess supply); any price below creates a shortage (excess demand). Market forces push the price back toward equilibrium.



Price Elasticity of Demand

Elasticity measures the responsiveness of quantity demanded to a change in price. It answers: "If price changes by 1%, by how much does quantity demanded change?"

Price Elasticity of Demand

$$E_d = \frac{\%\Delta Q_d}{\%\Delta P} = \frac{\partial Q}{\partial P} \cdot \frac{P}{Q}$$

Interpretation:

- $|E_d| > 1$: **Elastic** demand – quantity is highly responsive to price. A 1% price increase causes more than 1% drop in quantity.
- $|E_d| < 1$: **Inelastic** demand – quantity is not very responsive. Necessities (insulin, gasoline) tend to be inelastic.
- $|E_d| = 1$: **Unit elastic** – revenue is maximized at this point.

Elasticity and Revenue

If demand is **elastic** ($|E_d| > 1$): raising price $\Rightarrow$ total revenue *falls* (the quantity drop more than offsets the price increase).
If demand is **inelastic** ($|E_d| < 1$): raising price $\Rightarrow$ total revenue *rises*.
This is the most important managerial implication of elasticity.

Cross-Price and Income Elasticity

Other Elasticities

Cross-price elasticity: $E_{xy} = \frac{\%\Delta Q_x}{\%\Delta P_y}$

$E_{xy} > 0$: goods are *substitutes* $E_{xy} < 0$: goods are *complements*

Income elasticity: $E_I = \frac{\%\Delta Q_d}{\%\Delta I}$

$E_I > 0$: *normal good* $E_I < 0$: *inferior good* $E_I > 1$: *luxury good*

OLS Regression and Estimating Elasticity (FrescoBrew Case)

Managers rarely observe elasticity directly. OLS (Ordinary Least Squares) regression allows us to estimate it from data by isolating the effect of price on quantity while *controlling for* other demand shifters.

Why controls matter: Without controlling for confounders, the elasticity estimate is *biased*. For example, if ice cream sales rise in summer (demand shifter: temperature) and prices also rise in summer, a naïve regression of Q on P without temperature controls may show a *positive* or near-zero relationship—completely wrong.

OLS Regression Model for Demand

$$Q = \beta_0 + \beta_1 P + \beta_2 \text{Ad} + \beta_3 \text{Temp} + \beta_4 \ln(\text{Income}) + \beta_5 P_{\text{comp}} + \varepsilon$$

From β_1, compute point elasticity: $E_d = \beta_1 \cdot \frac{\bar{P}}{\bar{Q}}$

The FrescoBrew Three-Estimate Comparison

Method	What It Does	Typical Result
Naïve correlation	Regress Q on P only	Biased (too low $\lvert E_d \rvert$)
OLS with controls	Add demand shifters as controls	Better, but may still have bias
IV (Instrumental Variables)	Use supply-side cost shocks to isolate movements along the demand curve	Most credible estimate

The cost-shock variable belongs on the *supply* side. Including it in the demand equation "absorbs" the very price variation we need to trace the demand curve, biasing the price coefficient toward zero. Use it as an *instrument*, not a control.

IV Intuition: We want price changes that are driven only by supply shifts (cost shocks), because those trace out the demand curve. The IV approach: (1) regress P on cost shock and other variables; (2) construct predicted price from only the cost-shock component; (3) regress Q on predicted price and demand-side controls. The resulting elasticity reflects movement *along* the demand curve.

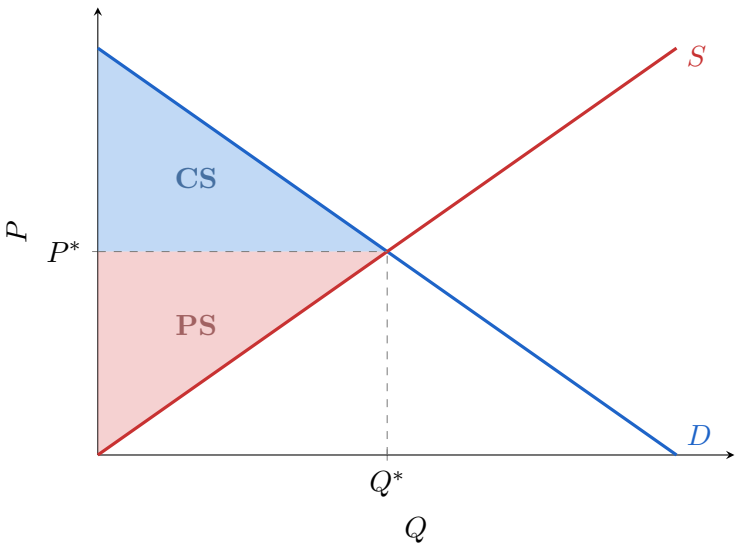
Module 3: Welfare, Price Controls, and Minimum Wage (Session 4)

Consumer Surplus, Producer Surplus, and Social Surplus

Consumer surplus (CS) is the difference between what consumers are willing to pay and what they actually pay. Graphically, it is the area below the demand curve and above the market price.

Producer surplus (PS) is the difference between the price sellers receive and the minimum they would accept. It is the area above the supply curve and below the market price.

Social surplus = CS + PS. It is maximized at the free-market equilibrium.



The Uber Case: Measuring Consumer Surplus

Uber's surge pricing provided a natural experiment. When Uber multiplied prices by 1.4×, some riders declined (revealing their willingness to pay was below the surge price). By measuring how the number of accepted rides changed at different surge multipliers, economists estimated the demand curve and found:

- Uber demand elasticity: $E_d \approx -0.6$ (inelastic)
- Consumer surplus: ≈ \$7 billion
- Total revenues (B+C): ≈ \$4 billion
- Producer surplus: ≈ \$1 billion

A **Pareto-efficient solution** to the Uber vs. taxi medallion conflict would use part of the \$7B consumer surplus generated to compensate medallion owners for their ≈\$1B loss, making no party worse off.

Price Ceilings and Price Floors

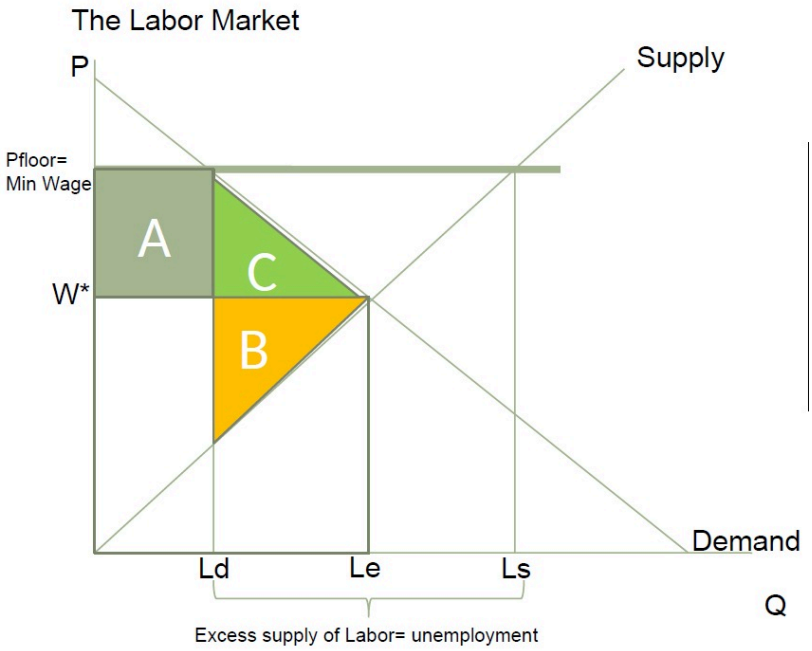
Price Ceiling (Maximum Price)

A price ceiling set *below* equilibrium creates a **shortage** (quantity demanded exceeds quantity supplied). Examples: rent control, Venezuelan price controls.

Price Floor (Minimum Price)

A price floor set *above* equilibrium creates a **surplus** (quantity supplied exceeds quantity demanded). Examples: minimum wage, EU agricultural price supports.

The Minimum Wage Debate



Reading the diagram:

- **A** = Gain to workers who keep their jobs at higher wage
- **B** = Loss to workers who lose their jobs
- **C** = Loss to firms (reduced producer surplus)
- **B+C** = Deadweight Loss (DWL) – social inefficiency
- **A−B** = Net effect on workers

The Elasticity Determines Who Wins

If labor demand is **inelastic** ($|E_d| < 1$): area A is large relative to B, so workers gain on net. Few jobs are lost.
If labor demand is **elastic** ($|E_d| > 1$): area B is large relative to A, so workers lose on net. Many jobs are lost.
Empirical estimate for low-skilled US labor: $E_d \approx -0.28$ (inelastic), but the elasticity of substitution between skilled and unskilled labor is about -1.5 (elastic), suggesting the long-run picture may be different.

UBI as an Alternative: A Universal Basic Income avoids the deadweight loss of minimum wages because it does not interfere with wage-setting. However, UBI may create labor supply disincentives (people choosing not to work) and requires substantial tax revenue to fund.

Module 4: Tax Incidence and the Laffer Curve (Session 5)

Average Effective Tax Rate (AETR)

Average Effective Tax Rate

$$\text{AETR} = \frac{\text{Tax Revenues}}{\text{GDP}}$$

AETR is **pro-cyclical**: it increases with GDP because tax revenues increase more than proportionally when GDP grows (progressive taxation).

Marginal vs. Average Tax Rates

Average tax rate = total taxes paid ÷ total income earned.
Marginal tax rate = tax rate on the *last* (marginal) unit of income earned.

In a **progressive** tax system, both rates increase with income, but the marginal rate rises faster. The *disincentive* effect of taxes depends primarily on the **marginal** rate, not the average rate.

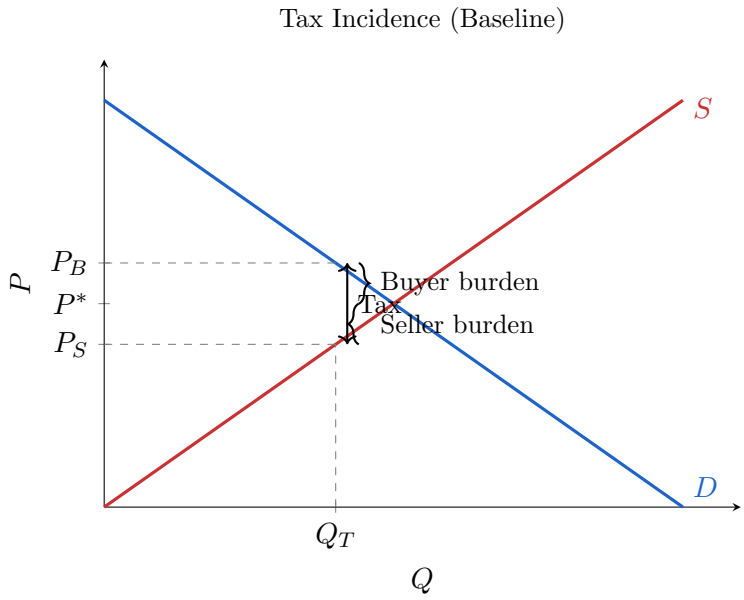
Spain has relatively low average tax rates but quite high marginal rates (exceeding 50% for annual incomes above 60,000). The true disincentive to work or invest is much larger than the average rate suggests.

Tax Incidence Theory

Tax incidence asks: *who actually bears the economic burden of a tax?* The answer depends not on who the tax is legally levied on, but on the relative elasticities of supply and demand.

Three key results:

1. Buyers and sellers **share** the burden of a tax.
2. The **inelastic** side of the market bears most of the burden.
3. The economic incidence is the same **regardless of** whether the tax is levied on buyers or sellers.

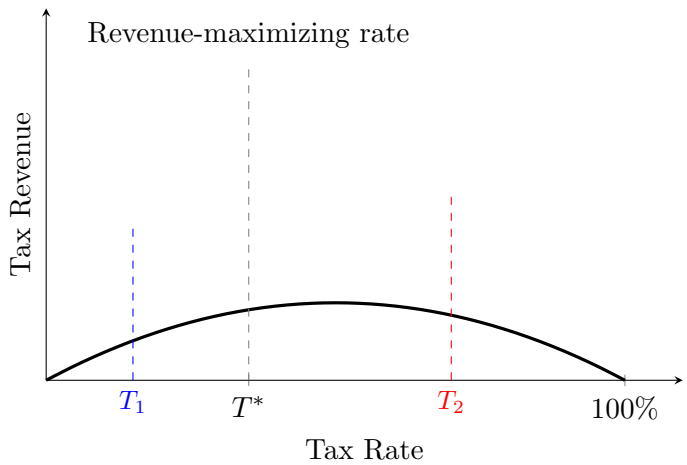


Application – US import tariffs: Tariffs on imported machinery are levied on US importers. Because export supply is relatively *elastic* and US demand for cars is relatively *inelastic*, US consumers bear most of the tariff burden. Research confirms: Americans pay almost entirely for Trump's tariffs.

Deadweight Loss (DWL) of Taxes

Every tax (except Pigovian taxes correcting externalities) creates a deadweight loss because it distorts incentives, reducing mutually beneficial trade. The DWL grows *more than proportionally* with the tax rate—doubling the tax rate more than doubles the DWL.

The Laffer Curve



Key Insight

The Laffer curve shows that there exist two different tax rates (T_1 and T_2) that generate the same revenue. T^* maximizes revenue, but this is *not necessarily the optimal tax rate* because the DWL at T^* may be very high. Tax cuts "pay for themselves" only if the current rate is above T^*. Research by Chodorow-Reich et al. (2024) found that Trump's 2017 corporate tax cuts reduced tax revenues by 40%, suggesting the US was **below** T^* for corporate taxes.

Canada's Carbon Tax + Income Rebate (CT+IR)

A **Pigovian tax** corrects a negative externality by making the private cost equal the social cost. Canada's carbon tax does this and rebates the revenue to households.

If all countries adopt CT+IR: The DWL from the externality is eliminated. Consumer welfare = CS (improved).

If only Canada adopts CT+IR: The climate change problem is not solved (one country acting alone has negligible effect). Consumers may be worse off because they pay the tax but get no environmental benefit. However, if sellers absorb part of the tax, the rebate may exceed consumers' direct tax burden, potentially making them better off on net.

Bottom line: Canada's CT+IR is theoretically attractive but remains a tough political sell because (1) it is not a global policy, (2) there is a lack of transparency and public awareness about the rebate, and (3) producers bear real costs (competitiveness, jobs).

Module 5: Measuring Economic Activity – GDP and CPI (Session 6)

Gross Domestic Product (GDP)

GDP is the **market value of all final goods and services produced within a country in a given period of time**. It was invented by Simon Kuznets in 1937.

GDP by Expenditure

$$\text{GDP} = C + I + G + NX$$

Where: C = Consumption, I = Investment, G = Government Purchases, NX = Net Exports (Exports − Imports)

What GDP excludes: home production, underground economy, non-market goods/services, illicit trades, expenditures on "bads," quality of environment/health/education, income inequality. GDP measures market value, not consumer surplus.

Nominal vs. Real GDP

Real GDP

$$\text{Real GDP}_t = \frac{\text{Nominal GDP}_t}{\text{GDP Deflator}_t} \times 100$$

The GDP deflator captures the change in the average level of prices. **Real GDP** isolates changes in actual output from changes in prices.

Peru's nominal GDP grew by up to 5000% in a single year during the hyperinflation of 1988–1990, yet real GDP collapsed. Always use **real** GDP for meaningful comparisons across time.

PPP-Adjusted GDP for Cross-Country Comparisons

Different countries have different price levels. A dollar buys more in India than in Switzerland. **Purchasing Power Parity (PPP)** adjusts for this by using an exchange rate that equalizes the cost of a standard basket of goods.

Big Mac Index (PPP Example)

$$e_{\text{PPP}} = \frac{P_{\text{Big Mac, China}}}{P_{\text{Big Mac, US}}} = \frac{25.50 \text{ yuan}}{6.01 \text{ USD}} = 4.24 \text{ yuan/\$}$$

Market rate: 7.19 yuan/\$. Implied undervaluation of yuan: $\frac{4.24-7.19}{7.19} \approx -41\%$

Rule of thumb: If GDP in PPP terms > GDP in current \$, the country's cost of living is *lower* than in the US, and its currency is *undervalued*.

Extensive vs. Intensive Growth

Extensive growth: Growth from more inputs (more workers, more capital). Cannot sustainably raise income per capita.

Intensive growth: Growth from higher **productivity** (output per worker). This is the *only* way to achieve long-term improvements in living standards.

China's economic slowdown is fundamentally a *productivity* slowdown. The "**Middle Income Trap**" (Eichengreen et al., 2013) refers to countries that stall around \$17,000 PPP per capita because they exhaust extensive growth without making the transition to productivity-driven intensive growth.

GDP, Growth, and Inequality – The GINI Coefficient

The GINI coefficient measures income inequality across an entire distribution (0 = perfect equality, 1 = perfect inequality). Globalization has *decreased* global inequality (between countries) but *increased* within-country inequality, especially in developed economies (the Lakner-Milanovic "elephant curve").

The Consumer Price Index (CPI)

Inflation Rate

$$\text{Inflation Rate}_t = \frac{\text{CPI}_t - \text{CPI}_{t-1}}{\text{CPI}_{t-1}} \times 100$$

CPI is a weighted average of prices: $\text{CPI} = \sum_i s_i \cdot P_i$, where s_i are expenditure share weights held **constant** from year to year.

Core vs. Headline inflation: Core CPI excludes food and energy (which are volatile). Headline CPI includes everything. Central banks often focus on core because it better reflects underlying trends.

CPI problems: (1) *Substitution bias* – the fixed basket doesn't account for consumers switching to cheaper alternatives; (2) *Unmeasured quality changes* and new goods – a smartphone today is far better than one from 2010 at the same price. These problems cause CPI to *overestimate* true inflation by roughly 1% per year.

PCE vs. CPI: The Personal Consumption Expenditures (PCE) index uses flexible weights (adjusts for substitution) and broader coverage (includes employer-paid healthcare). The Fed prefers PCE. Most years, PCE inflation is below CPI inflation.

Real vs. Nominal Interest Rates

Fisher Equation (Approximate)

$$r_{\text{real}} = r_{\text{nominal}} - \pi$$

where π is the inflation rate. The real interest rate is what matters for borrowing/saving decisions.

Inflating away debt: Governments can reduce real debt burdens through inflation. Rogoff documents that this is historically common after crises. However, with higher interest rates, this strategy becomes less viable as bondholders demand higher nominal returns.

Indexing and COLA: To convert dollar figures across time:

$$\text{Salary}_{2025} = \text{Salary}_{1950} \times \frac{\text{Price Level}_{2025}}{\text{Price Level}_{1950}}$$

Module 6: Fiscal and Monetary Policy (Sessions 7–8)

The AS-AD Framework

The **Aggregate Supply – Aggregate Demand** model is the central tool for macroeconomic diagnosis and prescription.

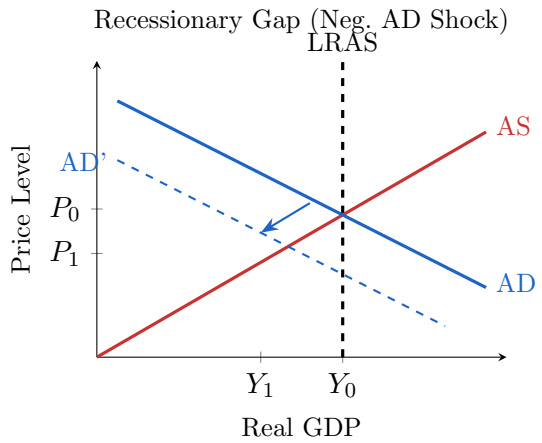
Aggregate Demand (AD) slopes downward: higher price levels reduce consumption (wealth effect), reduce investment (higher interest rates), and reduce net exports.

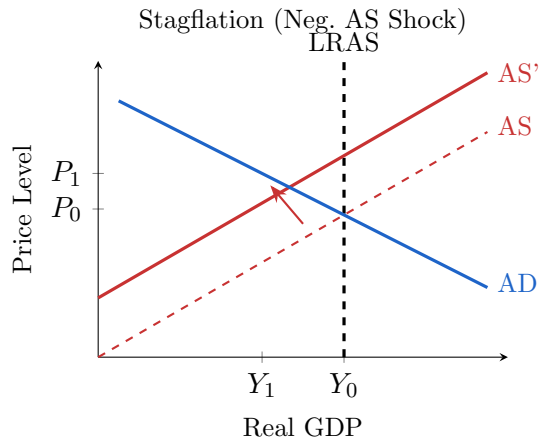
Short-run Aggregate Supply (AS) slopes upward: higher price levels increase production as firms earn higher margins.

Long-Run AS (LRAS) is vertical at potential GDP (Y_0): in the long run, output is determined by technology, labor, and capital, not by the price level.

Four Macroeconomic Scenarios

Scenario	Characteristics	Standard Policy
Recessionary Gap	GDP < potential; high unemployment; low inflation/deflation	Expansionary fiscal & monetary
Inflationary Gap	GDP > potential; low unemployment; rising inflation	Contractionary fiscal & monetary
Stagflation	High inflation + high unemployment (negative supply shock)	Supply-side reforms; no easy fix
Steady State	GDP ≈ potential; stable prices	Maintain current policy





The Phillips Curve

The Phillips curve illustrates the short-run tradeoff between inflation and unemployment. Expansionary policy moves the economy up and left (lower unemployment, higher inflation); contractionary policy moves it down and right.

The Phillips Curve Can Shift

In the 1970s, the Phillips curve shifted outward (up): both inflation and unemployment rose simultaneously (stagflation). This was caused by negative supply shocks (oil prices, collapse of the postwar labor accord). Supply-side reforms of the 1980s–90s (Reaganomics, Thatcherism) shifted it back inward, producing the "Great Moderation."

Policy Toolkit

Fiscal Policy

- **Expansionary:** Tax cuts and/or increased government spending ⇒ shifts AD right
- **Contractionary:** Tax increases and/or spending cuts ⇒ shifts AD left
- **Automatic stabilizers:** Tax revenues fall and welfare spending rises automatically in recessions, cushioning the downturn without explicit policy action

Monetary Policy

- **Expansionary:** Cut interest rates; Quantitative Easing (QE – central bank buys bonds) ⇒ shifts AD right
- **Contractionary:** Raise interest rates; Quantitative Tightening (QT – sell bonds) ⇒ shifts AD left

Arguments Against Active Stabilization Policy

General: (1) Automatic stabilizers already do much of the work; (2) Uncertainty about the right policy dose; (3) Policy lags (information, decision, implementation—each can take months to years).

Against fiscal policy specifically:

- **Ricardian Equivalence:** If consumers are forward-looking, they save tax cuts to pay for future tax increases, neutralizing the stimulus.
- **Crowding out:** Government borrowing raises interest rates, reducing private investment. $\Delta G > 0 \Rightarrow \Delta r > 0 \Rightarrow \Delta I < 0$.
- **Multiplier uncertainty:** Government purchases multipliers are estimated at 0.6–1.0; tax multipliers at -2 to -3 (Ramey, 2019), but estimates vary widely.
- **Debt accumulation:** Reinhart & Rogoff found that debt-to-GDP above 90% is associated with lower growth (though causality is debated).

Against monetary policy: Central banks don't control real interest rates; long and variable lags (2–3 years for peak impact); time-inconsistency problem.

Government Debt Dynamics

Debt Dynamics Equation

$$b_{t+1} = \frac{1 + r_t}{1 + g_t} \cdot b_t - s_t$$

Where: b_t = debt-to-GDP ratio, r_t = nominal interest rate, g_t = nominal GDP growth rate, s_t = primary surplus.

For a **constant** debt ratio ($b_{t+1} = b_t = b$):

$$s = \frac{r - g}{1 + g} \cdot b$$

Key insight: Debt is sustainable if nominal GDP growth (g) exceeds the interest rate (r). When $r > g$, a primary surplus is required.

Key definitions from "Things to teach yourself":

- **Primary deficit:** Government deficit excluding interest payments on debt
- **Structural deficit:** Deficit adjusted for the business cycle (a better measure of "true" fiscal stance)
- **Output gap:** Actual GDP minus potential GDP
- **Fiscal space:** A government's ability to take on more debt without jeopardizing stability
- **Fiscal dominance:** When debt levels are so high that monetary policy is constrained by fiscal needs

Historical Episodes

Episode	Type	Key Lesson
Great Depression (1930s)	Neg. AD shock	Markets don't self-correct; government must act. Gold standard prevented effective monetary policy. Roosevelt's New Deal changed expectations.
Golden Age (1948–73)	Steady growth	Keynesian demand management worked; larger government provided automatic stabilization.
Stagflation (1970s)	Neg. AS shock	Supply-side crises need supply-side solutions (Reaganomics). Excessive worker bargaining power + oil shocks.
Great Moderation (80s–00s)	Low inflation	Deregulation worked but created rising inequality and debt, setting stage for 2008 crisis.
Great Recession (2008)	Neg. AD shock	Excessive deregulation; revival of Keynes; massive QE + fiscal stimulus (ARRA).
COVID (2020)	Combined	AD collapse + AS disruption + LRAS shock; massive fiscal & monetary response; followed by inflation surge.

Module 7: International Trade and Tariffs (Session 9)

Comparative Advantage

David Ricardo's theory of **comparative advantage** states that countries benefit from trade by specializing in goods where they have the *lowest opportunity cost*, not the lowest absolute cost. Even if Country A is better at producing everything, both countries gain from specializing and trading.

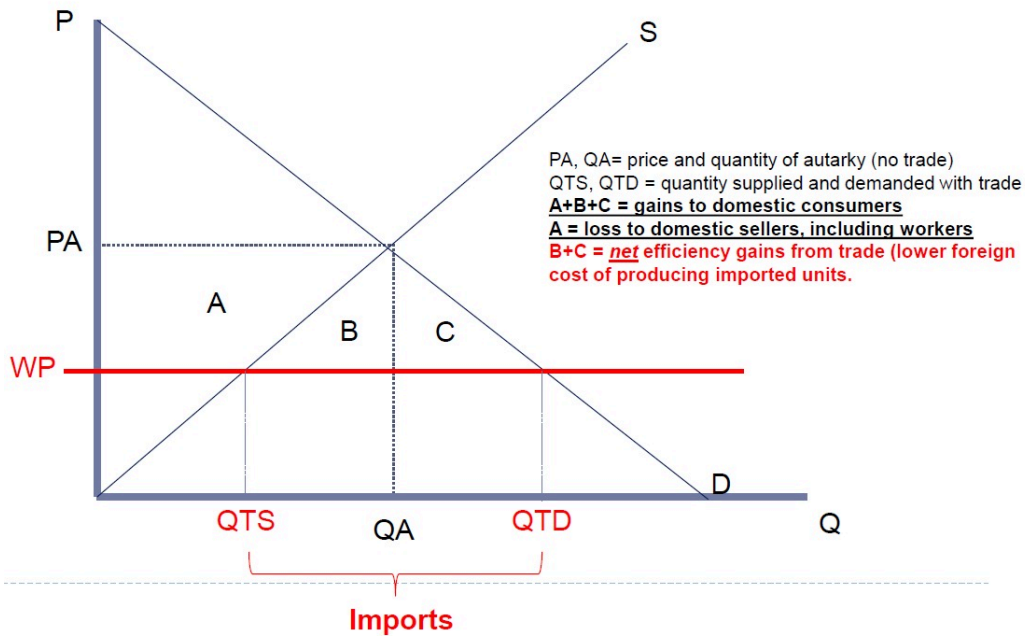
Opportunity Cost and Comparative Advantage

Opportunity cost of producing Good X = amount of Good Y sacrificed per unit of X produced.

A country has a **comparative advantage** in X if its opportunity cost of X is lower than the other country's.
A country has an **absolute advantage** in X if it can produce X with fewer resources.

Lucy and Bob example: Lucy takes 3h to hunt and 2h to find roots. Bob takes 4h and 5h. Lucy has absolute advantage in both. But Lucy's opportunity cost of hunting = 3/2 = 1.5 roots; Bob's = 4/5 = 0.8 roots. Bob has a comparative advantage in hunting. By specializing, total output increases 25% for Lucy and 12.5% for Bob.

Gains and Losses from Trade (Neoclassical Analysis)

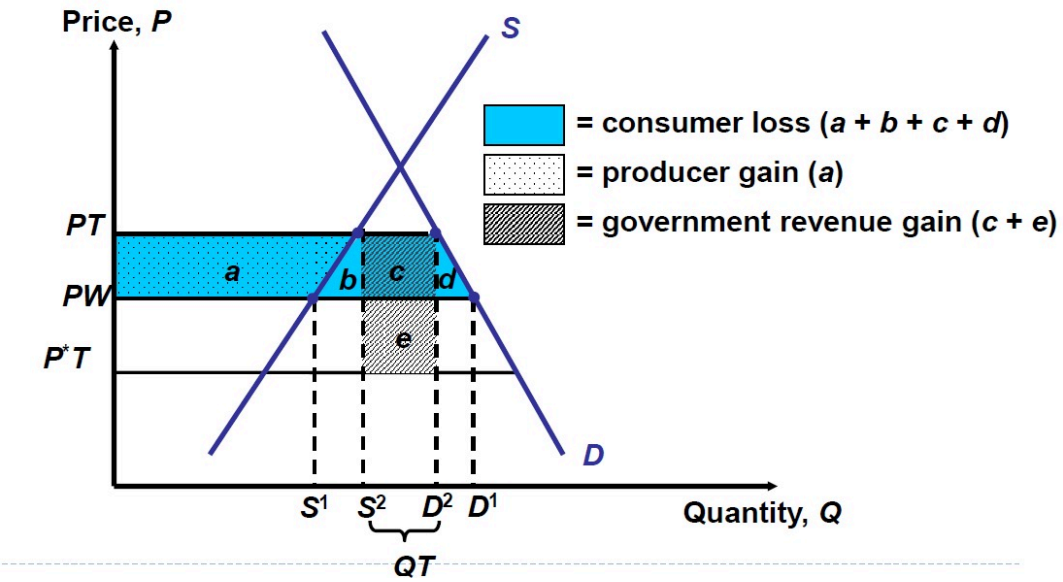


Under free trade at world price P_W: A+B+C = gain to domestic consumers; A = loss to domestic producers/workers; **B+C = net efficiency gain** from trade. A study found China's WTO accession lowered US prices by 7.6% in just five years.

The China Shock (Autor et al., 2016)

Trade with China (especially after 2001 WTO accession with Most Favored Nation status) led to: 8 million US manufacturing jobs lost since 1980; 3 million in just 6 years (2001–2007). The gains from trade were *diffused* across all consumers through lower prices, but the losses were *concentrated* in specific regions and industries (the "heartland"), creating significant political backlash.

Tariff Analysis



- a = Producer gain (domestic firms sell more at higher price)
- $b + d$ = Efficiency loss (deadweight loss) – production and consumption distortions
- $c + e$ = Government revenue
- e = Terms of trade gain (only for large countries that can affect world prices)
- **Net welfare** $= e - (b + d)$: positive only if terms of trade gain exceeds efficiency loss

Second-round effects are often more important: retaliation (Smoot-Hawley 1930 led to 60% decline in US trade), reduced macroeconomic discipline, and inflation.

Module 8: Industrial Policy in an Open Economy (Session 10)

The Infant Industry Argument

Under free trade, a dominant foreign industry (e.g., China in batteries) with economies of scale can produce at lower average cost, preventing domestic industry from ever developing. If protected by tariffs/subsidies, the domestic industry can grow, achieve scale economies, and eventually produce at *lower* cost than the foreign competitor. Free trade may be welfare-detrimental if it prevents this learning process.

Import Substituting Industrialization (ISI) vs. Export Oriented Industrialization (EOI)

	ISI	**EOI**
Focus	Replace imports with domestic production	Produce for export to global markets
Policy tools	High tariffs, quotas, subsidies	Trade openness, export incentives, competitive exchange rates
Competition	Low (protected domestic market)	High (global market discipline)
Typical outcome	Inefficiency, rent-seeking, balance-of-payments pressures	Higher productivity, export diversification, faster growth
Examples	Latin America until 1990s, East Asia until 1970s	South Korea, Taiwan, Singapore, China post-1970s

Economies of scale at the industry level: Firm size doesn't matter; spillovers between firms (knowledge, labor pooling, logistics); high competition; industry clusters. Main issue: coordination failure.

Economies of scale at the firm level: Firm size matters; learning by doing; oligopolistic competition; upstream/downstream spillovers. Main issue: credit constraints.

Problems with ISI: (1) Wrong targets (lobbying, lack of information); (2) Lack of competition breeds inefficiency; (3) Output growth $\neq$ success (can have large, inefficient employment); (4) Retaliation risk; (5) Dependency—high cost of quitting.

Module 9: Exchange Rates and Balance of Payments (Session 11)

Exchange Rate Regimes

Regime	Description and Examples
Dollarization / Currency board	Another country's currency circulates (Ecuador, Panama)
Supra-national	Common currency (Eurozone)
Pegged	Fixed to a reference currency with central bank intervention (Saudi Arabia to USD, Denmark to EUR)
Managed float	Government intervenes but no fixed target (China)
Independent float	Market-determined (USD, EUR, JPY, GBP)

The Balance of Payments

Balance of Payments Identity

$$\text{Current Account} + \text{Financial Account} = 0$$

Current Account (CA): Trade balance + services balance + net transfers
Financial Account (FA): Net foreign investment in the country (FDI + portfolio)

A CA deficit must be financed by a FA surplus (capital inflows), or the currency depreciates.
From the national accounts: $S - I = NX$, so a CA deficit means the country needs foreign capital.

CA deficit ⇒ country's net indebtedness increases ⇒ pressure for currency depreciation.
CA surplus ⇒ country lends abroad ⇒ pressure for currency appreciation.
FA surplus ⇒ foreign investment flowing in ⇒ pressure for currency appreciation.

Determinants of Exchange Rates

1. Inflation (Law of One Price / PPP):

Law of One Price

$$e_{A/B} = \frac{P_A}{P_B}$$

If Country A's inflation > Country B's, Country A's currency depreciates. Dramatically illustrated by the German hyperinflation (1921–25): prices rose 10^{13}-fold, and the Mark depreciated in proportion.

2. Monetary policy: Contractionary monetary policy (higher interest rates) tends to *appreciate* the currency by attracting capital inflows. But very high rates can also signal recession

risk, which is depreciating.

3. Fiscal policy: Expansionary fiscal policy tends to *appreciate* the currency (higher interest rates, higher expected growth attract investment). But it can also increase deficits and debt, which may depreciate long-term.

4. Trade policy: Import tariffs tend to reduce imports, improving the trade balance and appreciating the currency in the short run. But retaliation and reduced trade volume can offset this.

5. Safe-haven status: In crises, the USD and other safe-haven currencies appreciate regardless of fundamentals.

6. Central bank intervention: Buying or selling domestic currency using foreign exchange reserves. Rising reserves suggest the central bank is buying foreign currency to prevent appreciation (currency likely undervalued). Falling reserves suggest selling foreign currency to prevent depreciation (currency likely overvalued).

Fiscal and Monetary Policy Interactions with Exchange Rates

Key Policy-Exchange Rate Interactions

Monetary policy reinforces exchange rate effects: Higher interest rates ⇒ appreciation ⇒ both are contractionary.
Fiscal policy is partially offset: Expansionary fiscal ⇒ higher rates ⇒ appreciation ⇒ lower NX (offsets the stimulus).

Policy mix effects:
Expansionary fiscal + contractionary monetary ⇒ strong appreciation (US 2017–18)
Contractionary fiscal + expansionary monetary ⇒ depreciation (Eurozone 2014–15)
Both expansionary ⇒ ambiguous (US 2009–10)

The US Dollar Outlook

The US has run persistent CA deficits for 50+ years, financed by FA surpluses (the US is the top FDI destination globally). Whether the dollar appreciates or depreciates depends on whether capital inflows (attracted by growth, safety, technology leadership) offset trade deficits. Recent factors pointing toward USD depreciation: expansionary fiscal policy, persistent deficits, declining safe-haven status, sticky inflation.

China's sterilization: China accumulated enormous CA surpluses with the US. Normally this would appreciate the yuan, but the People's Bank of China "sterilized" the inflows by buying USD-denominated assets (US Treasuries), keeping the yuan artificially weak and the dollar artificially strong.

Additional Key Concepts (Professor's Exam Hints)

These concepts were specifically highlighted by Professor Fernández Kranz for exam preparation:

Concept	**Definition / Explanation**
Extensive vs. Intensive growth	Extensive = more inputs; Intensive = higher productivity. Only intensive growth raises living standards sustainably.
Primary deficit	Government deficit *excluding* interest payments on debt. Shows the "true" fiscal effort.
General government deficit	Total deficit including interest payments.
Structural deficit	Deficit adjusted for the business cycle (removes automatic stabilizer effects). Best measure of true fiscal stance.
Output gap	Actual GDP minus potential GDP. Positive = inflationary gap; negative = recessionary gap.
Inflationary gap	GDP above potential. Economy overheating; low unemployment; rising inflation.
Disinflation	Falling rate of inflation (prices still rising, but more slowly).
Deflation	Falling price level (negative inflation). Dangerous because it increases real debt burdens and delays consumption.
Stagflation	High inflation + high unemployment simultaneously. Caused by negative supply shocks.
Growth above/below potential	Above potential ⇒ unsustainable; inflationary pressures. Below ⇒ slack; unemployment.
Expansionary monetary policy	Lower interest rates and/or QE to stimulate demand.
Quantitative Easing (QE)	Central bank buys government bonds (and other assets) to inject money into the economy. Fed balance sheet grew 882% from 2008–2023.
Quantitative Tightening (QT)	Central bank sells bonds / lets them mature without reinvesting to reduce money supply.
Fiscal space	Government's room to increase spending or cut taxes without jeopardizing debt sustainability.
Fiscal dominance	When public debt is so large that monetary policy is effectively controlled by fiscal needs (central bank forced to keep rates low or monetize debt).
Inflating the debt away	Using inflation to reduce the real value of outstanding debt. Less viable when interest rates are high.

Quick Reference Formula Sheet

Microeconomics

Price Elasticity of Demand: $E_d = \frac{\%\Delta Q_d}{\%\Delta P} = \beta_1 \cdot \frac{P}{Q}$ ($|E_d| > 1$: elastic; < 1: inelastic)

Cross-Price Elasticity: $E_{xy} = \frac{\%\Delta Q_x}{\%\Delta P_y}$ (> 0: substitutes; < 0: complements)

Income Elasticity: $E_I = \frac{\%\Delta Q_d}{\%\Delta I}$ (> 0: normal; < 0: inferior; > 1: luxury)

Consumer Surplus: Area below demand curve, above price

Producer Surplus: Area above supply curve, below price

Social Surplus: $= CS + PS$ (maximized at equilibrium; any policy distortion reduces it)

Tax Incidence

Tax burden sharing: Inelastic side bears more burden, regardless of legal incidence

AETR: $= \frac{\text{Tax Revenue}}{\text{GDP}}$ (pro-cyclical)

Average tax rate: $= \frac{\text{Total tax paid}}{\text{Total income}}$ **Marginal tax rate:** Rate on the last unit of income

Macroeconomics

GDP: $Y = C + I + G + NX$

Real GDP: $= \frac{\text{Nominal GDP}}{\text{GDP Deflator}} \times 100$

Inflation: $\pi_t = \frac{\text{CPI}_t - \text{CPI}_{t-1}}{\text{CPI}_{t-1}} \times 100$

Fisher Equation: $r_{\text{real}} \approx r_{\text{nominal}} - \pi$

Debt Dynamics: $b_{t+1} = \frac{1+r}{1+g} \cdot b_t - s_t$ (Stable if: $s = \frac{r-g}{1+g} \cdot b$)

Dollar Conversion: $\text{Value}_{2025} = \text{Value}_{1950} \times \frac{\text{CPI}_{2025}}{\text{CPI}_{1950}}$

International Economics

Law of One Price: $e_{A/B} = P_A/P_B$

PPP Exchange Rate: Equalizes cost of basket across countries

Balance of Payments: $CA + FA = 0$ ($S - I = NX$)

Terms of Trade: $= P_X/P_M$ (price of exports / price of imports)

Tariff net welfare (large country): $= e - (b + d)$ (terms of trade gain minus efficiency loss)

Practice Problems with Worked Solutions

Problem 1: Elasticity and Revenue

A company sells 10,000 units at \$50 each. The estimated price elasticity of demand is -1.5. If the company raises price by 10%, what happens to revenue?

Solution:
$\%\Delta Q = E_d \times \%\Delta P = -1.5 \times 10\% = -15\%$
New $Q = 10{,}000 \times 0.85 = 8{,}500$; New $P = 50 \times 1.10 = \$55$
Old revenue $= 10{,}000 \times 50 = \$500{,}000$
New revenue $= 8{,}500 \times 55 = \$467{,}500$
Revenue **falls** by \$32,500 (6.5%). Because demand is elastic ($|E_d| > 1$), the quantity drop more than offsets the price increase.

Problem 2: Tax Incidence

The government imposes a \$10 tax on sellers in a market where demand is relatively inelastic and supply is relatively elastic. Who bears most of the burden?

Solution:
By tax incidence theory, the **inelastic side** bears most of the burden. Since demand is more inelastic than supply, **buyers** bear most of the \$10 tax, even though it is legally levied on sellers. The price paid by buyers rises by more than \$5, and the price received by sellers falls by less than \$5.

Problem 3: Real vs. Nominal GDP

Country X's nominal GDP grew from \$500B to \$550B (10% growth). The GDP deflator increased from 100 to 106. What was real GDP growth?

Solution:
Real GDP in Year 2: $\frac{550}{106} \times 100 = \$518.9\text{B}$
Real GDP growth: $\frac{518.9-500}{500} \times 100 = 3.8\%$
Of the 10% nominal growth, 6% was inflation and only 3.8% was real output growth.

Problem 4: Debt Sustainability

Japan has a debt-to-GDP ratio of 240%, a primary deficit of 3.5% of GDP, and inflation of 2%. What nominal GDP growth rate is needed to stabilize the debt ratio at a nominal interest rate of 2%?

Solution:
Using $g = \frac{b \cdot r + d}{b - d}$ where $b = 2.40$, $r = 0.02$, $d = 0.035$:
$g = \frac{2.40 \times 0.02 + 0.035}{2.40 - 0.035} = \frac{0.048 + 0.035}{2.365} = \frac{0.083}{2.365} = 0.0351 = 3.51\%$
Real GDP growth needed: $3.51\% - 2\% = 1.51\%$. This is ambitious for Japan given its aging population and productivity challenges.

Problem 5: Comparative Advantage

Country A can produce 1 car in 100 hours or 1 ton of wheat in 10 hours. Country B can produce 1 car in 200 hours or 1 ton of wheat in 25 hours. Which country has a comparative advantage in cars?

Solution:
Country A: Opportunity cost of 1 car = 100/10 = 10 tons of wheat.
Country B: Opportunity cost of 1 car = 200/25 = 8 tons of wheat.
Country B has the **comparative advantage in cars** (lower opportunity cost: 8 vs. 10 tons of wheat), even though Country A has the absolute advantage in both goods. Country A should specialize in wheat; Country B in cars.

Problem 6: AS-AD Diagnosis

An economy experiences rising inflation and falling unemployment simultaneously. Diagnose the situation and prescribe policy.

Solution:
This is an **inflationary gap** (positive AD shock). GDP is above potential, pushing up prices while reducing unemployment. **Prescription:** contractionary monetary policy (raise interest rates, QT) and/or contractionary fiscal policy (reduce government spending, raise taxes) to shift AD left, bringing GDP back toward potential and reducing inflationary pressure. The tradeoff: unemployment will temporarily rise.

Problem 7: Exchange Rate Analysis

The US runs a large current account deficit and a large financial account surplus. The Fed raises interest rates. What happens to the dollar?

Solution:
The CA deficit puts *depreciating* pressure on the dollar. However, the FA surplus (capital inflows attracted by US growth and safety) puts *appreciating* pressure that has historically outweighed the CA deficit. The Fed rate hike further *appreciates* the dollar by attracting more capital (higher returns on USD assets). Net effect: dollar likely **appreciates** in the short run, though the persistent CA deficit remains a long-term depreciation risk.

End of Study Guide